

1. Write code to demo
 1. **singleton pattern** (both lazy loading and eager loading),
 2. **factory method pattern**
 3. **Abstract factory pattern**
 4. **Builder pattern**
5. PLEASE **DO NOT** USE DEMO CODE USED IN CLASS SESSIONS, PLEASE WRITE YOUR OWN CODE.
6. Write necessary POJOs together with your core implementation, for your **singleton pattern** implementation, explain how you would guarantee it is **thread-safe** (thread-safe in this scenario means: there is **truly only one** instance in JVM).
2. Write code to explain how do **default** and **static** keywords work in interfaces since Java 8
3. Write code to demo **Java anonymous class**, you may write your own POJOs (e.g. parent abstract classes, child classes, and interfaces)
4. Write code to explain **Lambda expression** with your own **functional interface**.
5. Write a calculator with **BiFunction<T,U,R>** (an internal functional interface provided by JDK) and **Lambda expression**. Your calculator should support two-number **addition, subtraction, multiplication, division** operations.